YEAR 11 PHYSICS, AQA GCSE, LESSON 1 OF 10

Atoms & Nuclear Radiation.

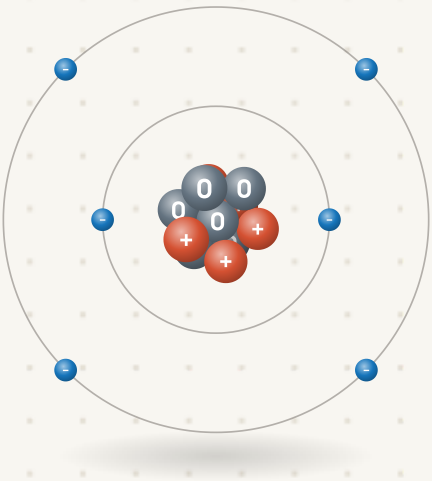
Lesson 01 Inside the atom

Everything around you is made of atoms, yet an atom is almost entirely empty space. So what is actually in there?

IN THIS LESSON YOU WILL

- describe the charge, mass and position of protons, neutrons and electrons
- state the size of an atom and of its nucleus
- explain why an atom has no overall charge
- use atomic and mass numbers to find the numbers of each particle

ANSWER EDITION FOR TEACHERS AND SELF-MARKING



NAME _____ CLASS _____

TEACHER _____ TARGET GRADE _____

START HERE

How this booklet works.

This unit starts from the atoms you met in Key Stage 3 and in chemistry, then asks how we know what is inside them and what happens when a nucleus is unstable. Keep it: it is your classwork book *and* your revision guide.

DO NOW
Quick questions from earlier lessons. Remembering again is what makes it stick.

LEARN IT
Explanations, diagrams and worked examples. Read this again when you revise.

TRY IT
Practice that builds up: first with support, then on your own, then a challenge.

LAB
Record what you find in a simulation or a model experiment.

EXAM CORNER
AQA-style questions with a tip on how the marks are given.

REVISE IT
Summary, cover-and-answer questions and an "I can" check to colour in.

! **Revising at home?** (1) Read the Learn it pages. (2) Cover them and answer the "Cover the page" questions in Revise it. (3) Check with the quick answers at the back. (4) Go through the matching slides in the teaching deck, and try the simulations.

The route through the unit (this lesson is highlighted)

01 **Inside the atom** Protons, neutrons and electrons, the size of an atom, atomic and mass numbers (slides 3–6)

02 Isotopes and ions

03 From plum pudding to the nucleus

04 Energy levels and the nucleus

05 Radioactive decay

06 Nuclear equations

07 Properties and uses of radiation

08 Half-life and random decay

09 Half-life calculations

10 Contamination and irradiation

Teaching deck: **panphy.app/gcsephy/decks/atoms-and-radiation**. Simulations: **panphy.app/simulations/atomic_models.html** and **panphy.app/simulations/nuclear_decay.html**

LESSON

01

Inside the atom.

AQA Physics 4.4.1.1–4.4.1.2; Combined Science 6.4.1.1–6.4.1.2

BIG QUESTION

Everything around you is made of atoms, yet an atom is almost entirely empty space. So what is actually in there?

BY THE END YOU CAN

- describe the charge, mass and position of protons, neutrons and electrons
- state the size of an atom and of its nucleus
- explain why an atom has no overall charge
- use atomic and mass numbers to find the numbers of each particle

atom · nucleus · proton · neutron · electron · atomic number · mass number

SLIDES AND SIMULATIONS

Teaching deck, slides 3–6

Three particles, the size of an atom, and reading the symbol.

panphy.app/gcsephy/decks/atoms-and-radiation/#3

RETRIEVE

What do you already know?

From Key Stage 3 and chemistry. Have a go.

1
What is everything around you made of?

Atoms (tiny particles).

2
Name the three particles found in atoms.

Protons, neutrons and electrons.

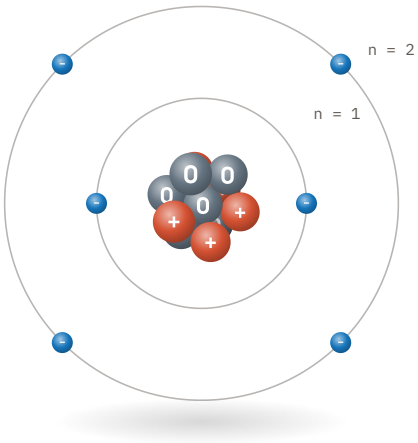
3
What is an element?

A substance made of only one type of atom.

EXPLAIN

What is inside an atom?

Three particles



A carbon atom: 6 protons and 6 neutrons in the nucleus, 6 electrons around it.

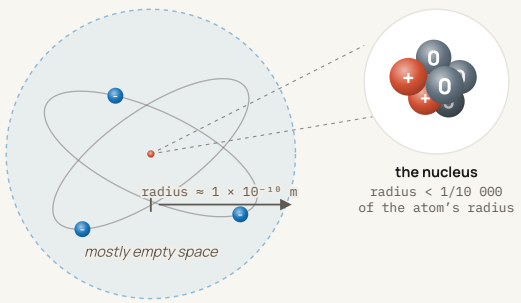
At the centre of every atom is a tiny **nucleus**. It contains **protons**, which are positive, and **neutrons**, which have no charge. So the nucleus is positive.

Negative **electrons** are arranged at different distances from the nucleus, called **energy levels**.

Particle	Relative charge	Relative mass	Where it is
Proton	+1	1	in the nucleus
Neutron	0	1	in the nucleus
Electron	−1	very small	in energy levels around the nucleus

! An atom has **no overall charge**: it has the **same number of electrons as protons**, so the positive and negative charges cancel. Neutrons have no charge, so they do not count.

Mostly empty space



Not to scale: a real nucleus is far too small to draw.

An atom has a radius of about $\mathbf{1 \times 10^{-10}}$ **m** (0.000 000 000 1 m).

The radius of the nucleus is **less than 1/10 000** of the radius of the atom. Yet **almost all the mass** of the atom is in the nucleus, because protons and neutrons are far heavier than electrons.

THINK OF A FOOTBALL STADIUM

If an atom were as big as a football stadium, the nucleus would be about the size of a pea on the centre spot. The electrons would be somewhere in the stands. Everything in between is empty space.

Where the model breaks: a pea is light, but the nucleus holds almost all the mass; and electrons are not little balls sitting in seats.

WORKED EXAMPLE

An atom has a radius of 1×10^{-10} m. Its nucleus is 1/10 000 of this. Calculate the radius of the nucleus.

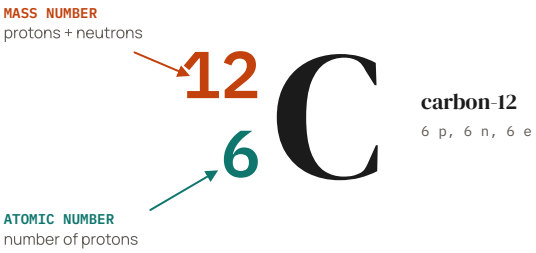
KNOW	atom radius = 1×10^{-10} m
METHOD	divide by 10 000 = 1×10^{4}
WORKING	$1 \times 10^{-10} \div 1 \times 10^{4}$
ANSWER	**1×10^{-14} m**

YOUR TURN

An atom has a radius of 3×10^{-10} m. Its nucleus is 1/10 000 of this. Calculate the radius of the nucleus.

KNOW	atom radius = 3×10^{-10} m
METHOD	divide by 1×10^{4}
WORKING	$3 \times 10^{-10} \div 1 \times 10^{4}$
ANSWER	3×10^{-14} m

Reading the symbol



- The **atomic number** (bottom) is the number of **protons**. It decides which element the atom is.
- The **mass number** (top) is the number of **protons + neutrons**.
- Number of neutrons = mass number − atomic number.
- In an atom, number of electrons = number of protons.

WORKED EXAMPLE

How many protons, neutrons and electrons are in an atom of $^{23}_{11}$Na?

PROTONS	atomic number = 11
NEUTRONS	23 − 11 = 12
ELECTRONS	same as protons = 11
ANSWER	**11 p, 12 n, 11 e**

YOUR TURN

How many protons, neutrons and electrons are in an atom of $^{27}_{13}$Al?

PROTONS	13
NEUTRONS	27 − 13 = 14
ELECTRONS	13
ANSWER	13 p, 14 n, 13 e

PRACTISE

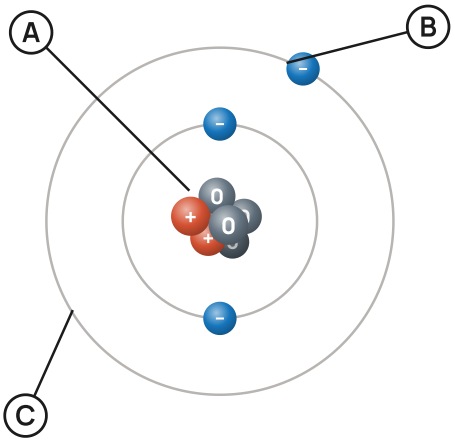
Try it

1 Complete the sentences using the word bank.

At the centre of an atom is a tiny **nucleus**. It contains **protons** and **neutrons**. **Electrons** are arranged in energy levels around it. A proton has a relative charge of **+1** and an electron has a relative charge of **−1**. Almost all of the **mass** of the atom is in the nucleus.

WORD BANK: nucleus | protons | neutrons | Electrons | mass | +1 | −1 | 0 | charge

2 The diagram shows a lithium atom. Name the parts labelled A, B and C. Then say which particles are inside part A.



A **nucleus**

B **electron**

C **energy level**

Inside A: **3 protons, 4 neutrons**

3 SUPPORTED Complete the table.

Symbol	Protons	Neutrons	Electrons
$^{4}_{2}He$	2	2	2
$^{7}_{3}Li$	3	4	3
$^{16}_{8}O$	8	8	8
$^{56}_{26}Fe$	26	30	26
$^{238}_{92}U$	92	146	92

4 ON YOUR OWN An atom has 17 protons and 18 neutrons. Write its symbol in the form $^{A}_{Z}X$. The element with 17 protons is chlorine, Cl.

$^{35}_{17}Cl$ (mass number 17 + 18 = 35)

5 Which part of an atom contains almost all of its mass? Tick **one** box.

- ☐ the electrons
- ☑ the nucleus
- ☐ the energy levels
- ☐ the empty space

6 Explain what is wrong with Leo's statement, and correct it.

> **Leo:** An atom is neutral because it has the same number of protons and neutrons.

Neutrons have no charge, so they do not balance anything. An atom is neutral because it has the same number of electrons (−) as protons (+), so the charges cancel.

7 CHALLENGE The radius of an atom is about 10 000 times the radius of its nucleus. If you drew the nucleus as a dot 1 mm across, how wide would you have to draw the atom? Give your answer in metres, and say whether it would fit on this page.

10 000 × 1 mm = 10 000 mm = 10 m. It would not fit on the page: this is why atom diagrams are never to scale.

AQA-STYLE

Exam corner

Write in full sentences. Show every step of a calculation.

Get the marks: A 'describe' question about structure needs where each particle is, not just its name.

8 Describe the structure of an atom. **[4 marks]**

A tiny nucleus at the centre (1), containing protons and neutrons (1). Electrons are at different distances from the nucleus / in energy levels around it (1). Most of the atom is empty space: the nucleus radius is less than 1/10 000 of the atom's (1).

9 An atom of potassium is $^{39}_{19}\mathrm{K}$. (a) Give the number of neutrons. (b) Explain why the atom has no overall charge. **[3 marks]**

(a) 39 − 19 = 20 (1). (b) It has 19 protons and 19 electrons (1); their charges are equal and opposite, so they cancel (1).

REVIEW LATER

Revise it

Use this box after the lesson and before tests.

SUMMARY

- The nucleus contains protons (+1) and neutrons (0). Electrons (−1) are in energy levels around it.
- Atom radius ≈ 1×10^{-10} m; nucleus radius less than 1/10 000 of this.
- Almost all the mass is in the nucleus.
- Atomic number = protons. Mass number = protons + neutrons.
- In an atom, electrons = protons, so there is no overall charge.

COVER THE PAGE AND ANSWER

1 What is the relative charge of a neutron?

2 What is the radius of an atom, roughly?

3 What does the mass number tell you?

4 How many neutrons are in $^{31}_{15}\mathbf{P}$?

I CAN…

- describe the charge, mass and position of each particle
- state the size of an atom and of its nucleus
- explain why an atom is neutral
- work out protons, neutrons and electrons from nuclear notation

REFERENCE

Toolkit.

The facts and methods you have met so far. Learn these: exam questions expect you to recall them without a data sheet.

The three particles

Lesson 1

Particle	Charge	Mass	Where
Proton	+1	1	nucleus
Neutron	0	1	nucleus
Electron	−1	very small	energy levels

Reading the symbol

Lesson 1

$^{A}_{Z}X$ e.g. $^{23}_{11}Na$

- Z, atomic number = protons
- A, mass number = protons + neutrons
- neutrons = A − Z; in an atom, electrons = Z
- Atom radius ≈ 1×10^{-10} m; nucleus radius < 1/10 000 of that

REFERENCE

Key words from this lesson.

Learn the meaning and spelling. In exams, precise words earn marks: say “nucleus”, not “middle”, and “unstable nucleus”, not “unstable atom”.

Atom: The smallest part of an element. It has a tiny nucleus with electrons around it and no overall charge.

Atomic number: The number of protons in the nucleus. It decides the element.

Electron: A tiny particle with a relative charge of −1, found in energy levels around the nucleus.

Mass number: The total number of protons and neutrons in the nucleus.

Neutron: A particle in the nucleus with no charge and a relative mass of 1.

Nucleus: The tiny centre of an atom, containing protons and neutrons and almost all the mass.

Proton: A particle in the nucleus with a relative charge of +1 and a relative mass of 1.

CHECK YOURSELF

Quick answers.

Answers to the “Do now” and “Cover the page” questions, so you can check yourself when revising. Answers to the longer questions are in the answer edition.

DO NOW

1. Atoms (tiny particles).
2. Protons, neutrons and electrons.
3. A substance made of only one type of atom.

COVER THE PAGE

1. 0
2. 1×10^{-10} m
3. The number of protons + neutrons.
4. 31 − 15 = 16